

Integrality of $\Gamma_H(G)$ forces H to be normal

Statement and source problem

Let G be a finite group of order $n \geq 5$, and let H be a proper, nontrivial subgroup of G . Following Definition 1.4 and the Section 2 subgroup convention of Poddar, Biswas, and Das, set

$$S_H = \{(g, 1), (1, g), (g, g) : g \in G \setminus H\}$$

and

$$\Gamma_H(G) = \text{Cay}(G \times G, S_H).$$

Thus the vertices are the elements of $G \times G$, and distinct vertices u, v are adjacent precisely when $uv^{-1} \in S_H$.

The source proves in Theorem 3.1 that $H \triangleleft G$ implies that $\Gamma_H(G)$ is integral. Its first bullet in Section 6, “Conclusion and Open Issues,” asks whether normality is also necessary. That bullet is unnumbered in the source; it is not assigned a conjecture number.

Integral means that every adjacency eigenvalue is an integer. We prove the converse.

Theorem. The graph $\Gamma_H(G)$ is integral if and only if $H \triangleleft G$.

Proof idea

Assume that H is not normal. The permutation representation of G on the cosets G/H must then contain an irreducible representation (ρ, V) whose H -fixed subspace is neither zero nor all of V . This representation detects the failure of normality.

The adjacency operator of a Cayley graph splits into blocks indexed by the irreducible representations of the underlying group. We use the block $\rho \otimes \rho^*$ of $G \times G$. After identifying $V \otimes V^*$ with $\text{End}(V)$, that block preserves the two-dimensional space spanned by the orthogonal projection onto V^H and its complementary projection. The characteristic polynomial on this plane has a discriminant equal to a rational square factor times a positive nonsquare integer. It therefore contributes two irrational adjacency eigenvalues.

1. An irreducible representation detecting nonnormality

We prove the contrapositive, so suppose that H is not normal. Use the source notation

$$n = |G|, \quad k = |H|, \quad \ell = [G : H],$$

so that $n = k\ell$.

Consider the complex permutation module on the left cosets,

$$\mathbb{C}[G/H] = \text{Ind}_H^G(\mathbf{1}_H).$$

For $(\sigma, W) \in \text{IRR}(G)$, Frobenius reciprocity says that the multiplicity of σ in this module is

$$\dim W^H.$$

Suppose that every irreducible constituent of $\mathbb{C}[G/H]$ were fixed pointwise by H . Then H would act trivially on the entire permutation module and hence on every coset. Thus, for all $a \in H$ and $g \in G$,

$$agH = gH.$$

Equivalently, $g^{-1}ag \in H$. It would follow that $g^{-1}Hg \subseteq H$; equality of orders would give $g^{-1}Hg = H$ for every g , contrary to the nonnormality of H .

Consequently there is an irreducible representation $(\rho, V) \in \text{IRR}(G)$ for which

$$0 < r := \dim V^H < d := \deg(\rho) = \dim V. \quad (1)$$

In particular, ρ is nontrivial and $d \geq 2$.

Taking dimensions in the irreducible decomposition of $\mathbb{C}[G/H]$ gives

$$\ell = \sum_{(\sigma, W) \in \text{IRR}(G)} (\dim W)(\dim W^H).$$

The trivial representation contributes 1, while the chosen representation contributes dr . Every other term is nonnegative, so

$$dr \leq \ell - 1, \quad \text{hence} \quad \ell \geq dr + 1. \quad (2)$$

Choose a G -invariant Hermitian inner product on V , making ρ unitary. In the source notation for sums of representation matrices, put

$$P = P_H := \frac{1}{k} \rho(H) = \frac{1}{k} \sum_{a \in H} \rho(a), \quad Q = I - P.$$

Then P is the orthogonal projection onto V^H , and therefore

$$P^2 = P = P^*, \quad \text{Tr}(P) = r. \quad (3)$$

By (1), both P and Q are nonzero.

2. A two-dimensional adjacency block

The contragredient (dual) representation ρ^* is irreducible, so

$$\Omega := \rho \otimes \rho^* \in \text{IRR}(G \times G).$$

Identify $V \otimes V^*$ with $\text{End}(V)$. Under this identification,

$$\Omega(a, b)X = \rho(a)X\rho(b)^{-1} \quad (a, b \in G, X \in \text{End}(V)). \quad (4)$$

The regular representation of $G \times G$ contains every irreducible representation. Hence the adjacency matrix of $\Gamma_H(G)$ contains the Fourier block

$$B := \Omega(S_H) = \sum_{s \in S_H} \Omega(s).$$

Explicitly,

$$B(X) = \sum_{g \in G \setminus H} (\rho(g)X + X\rho(g)^{-1} + \rho(g)X\rho(g)^{-1}). \quad (5)$$

Every eigenvalue of B is therefore an adjacency eigenvalue of $\Gamma_H(G)$.

We now evaluate B on P and Q . Since ρ is nontrivial and irreducible,

$$\sum_{g \in G} \rho(g) = 0. \quad (6)$$

Schur averaging on $\text{End}(V)$ gives

$$\sum_{g \in G} \rho(g) X \rho(g)^{-1} = \frac{n}{d} \text{Tr}(X) I. \quad (7)$$

Also, inversion permutes both H and $G \setminus H$, so

$$\sum_{g \in G \setminus H} \rho(g) = \sum_{g \in G \setminus H} \rho(g)^{-1} = -kP. \quad (8)$$

Finally, P and Q commute with every $\rho(a)$, $a \in H$. Subtracting the H -conjugation average from (7) and using (8) in (5) yields

$$B(P) = \left(\frac{nr}{d} - 3k \right) P + \frac{nr}{d} Q, \quad (9)$$

and

$$B(Q) = \frac{n(d-r)}{d} P + \left(\frac{n(d-r)}{d} - k \right) Q. \quad (10)$$

Thus $\langle P, Q \rangle$ is a two-dimensional B -invariant subspace. Set

$$\alpha = \frac{nr}{d}, \quad \beta = \frac{n(d-r)}{d}.$$

In the ordered basis (P, Q) , the restricted operator has matrix

$$\begin{pmatrix} \alpha - 3k & \beta \\ \alpha & \beta - k \end{pmatrix}. \quad (11)$$

Its trace and determinant are

$$\text{Tr} = n - 4k, \quad \det = 3k^2 - 3nk + \frac{2nkr}{d}. \quad (12)$$

Its discriminant is therefore

$$\begin{aligned} \Delta &= (n - 4k)^2 - 4 \left(3k^2 - 3nk + \frac{2nkr}{d} \right) \\ &= (n + 2k)^2 - \frac{8nkr}{d} \\ &= \left(\frac{k}{d} \right)^2 M, \end{aligned} \quad (13)$$

where, since $n = k\ell$,

$$M = d^2(\ell + 2)^2 - 8\ell dr. \quad (14)$$

3. The integer M is not a square

Define

$$N = d(\ell + 2) - 4r.$$

A direct expansion of (14) gives

$$M = N^2 + 16r(d - r). \quad (15)$$

By (1), $M > N^2$. Moreover, (2) implies

$$N \geq d(dr + 3) - 4r = r(d^2 - 4) + 3d > 0. \quad (16)$$

Except at two smallest parameter triples, M lies strictly between N^2 and $(N + 1)^2$; the two exceptions will be bracketed separately. For the general cases, the relevant gap is

$$(N + 1)^2 - M = 2d(\ell + 2) - 8r + 1 - 16r(d - r). \quad (17)$$

For fixed d, r , this expression increases by $2d$ when ℓ increases by 1. At the smallest value allowed by (2), $\ell = dr + 1$, it becomes

$$F(d, r) = 2rd(d - 8) + 16r^2 - 8r + 6d + 1. \quad (18)$$

For $d \geq 8$, all three terms

$$2rd(d - 8), \quad 16r^2 - 8r, \quad 6d + 1$$

are nonnegative, and the last two are positive. For $4 \leq d \leq 7$, the same positivity follows from the following forms:

d	$F(d, r)$
4	$(4r - 5)^2$
5	$16(r - 1)^2 - 6(r - 1) + 9$
6	$16(r - 1)^2 + 21$
7	$16(r - 1)^2 + 10(r - 1) + 37$

For $d = 5$, put $t = r - 1 \geq 0$. At $t = 0$ the expression is 9; at integer $t \geq 1$, $16t^2 - 6t + 9 \geq 10t + 9 > 0$. These expressions are positive for every integer $1 \leq r \leq d - 1$. The remaining case $d = 3, r = 2$ gives $F(3, 2) = 7 > 0$. In all these cases, (15), (17), and (18) imply

$$N^2 < M < (N + 1)^2. \quad (19)$$

There are only two exceptional smallest triples.

- If $(d, r, \ell) = (3, 1, 4)$, then $N = 14$ and $M = 228$, so

$$15^2 = 225 < 228 < 256 = 16^2.$$

For every $\ell \geq 5$, the expression in (17) is $-3 + 6(\ell - 4) > 0$.

- If $(d, r, \ell) = (2, 1, 3)$, then $N = 6$ and $M = 52$, so

$$7^2 = 49 < 52 < 64 = 8^2.$$

For every $\ell \geq 4$, the expression in (17) is $-3 + 4(\ell - 3) > 0$.

This exhausts $d \geq 2$ and $1 \leq r \leq d - 1$. Hence M is a positive integer that is not an integer square. An integer that is a square in $\mathbb{Q}$ is already an integer square, so $\sqrt{M} \notin \mathbb{Q}$.

4. Irrational adjacency eigenvalues

By (12)–(14), the two eigenvalues of the restriction (11) are

$$\lambda_{\pm} = \frac{n - 4k \pm (k/d)\sqrt{M}}{2}. \quad (20)$$

Both are irrational. Since this restriction belongs to the genuine $\Omega = \rho \otimes \rho^*$ adjacency block, both eigenvalues occur in $sp(\Gamma_H(G))$. Thus $\Gamma_H(G)$ is not integral whenever H is not normal.

This proves the contrapositive: if $\Gamma_H(G)$ is integral, then $H \triangleleft G$. Together with Theorem 3.1 of the source, the equivalence follows. $\square$

Dependencies and scope

The proof uses complete reducibility for finite groups over $\mathbb{C}$, Frobenius reciprocity, unitary realization of finite-group representations, the irreducibility of tensor products for direct products, the regular-representation decomposition of a Cayley adjacency operator, and Schur averaging. It uses no classification of finite groups and no finite enumeration. The assumptions on G , H , and $\Gamma_H(G)$ are those in Definition 1.4, the subgroup convention of Section 2, and the first open issue of Section 6.

Source

S. Poddar, S. Biswas, and A. Das, “An integral family of quasi-strongly regular Cayley graphs,” [arXiv:2511.14496v1](#). The graph $\Gamma_H(G)$ is Definition 1.4, the implication $H \triangleleft G \Rightarrow \Gamma_H(G)$ integral is Theorem 3.1, and the converse proved above is the first unnumbered open issue in Section 6.